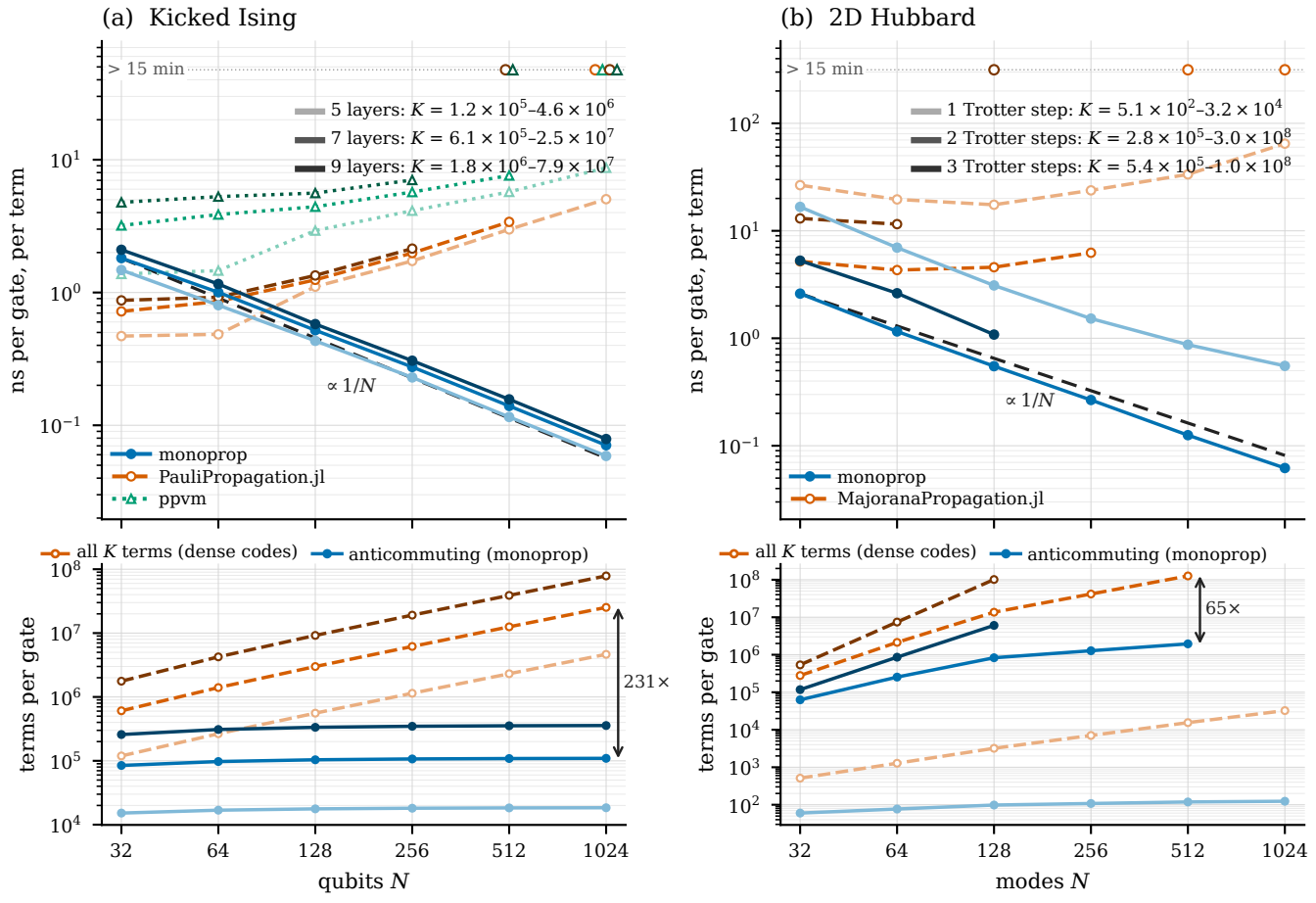
(a) Kicked Ising
(b) 2D Hubbard
> 15 min
5 layers: $K = 1.2 \times 10^5$–$4.6 \times 10^6$
7 layers: $K = 6.1 \times 10^5$–$2.5 \times 10^7$
9 layers: $K = 1.8 \times 10^6$–$7.9 \times 10^7$
1 Trotter step: $K = 5.1 \times 10^2$–$3.2 \times 10^4$
2 Trotter steps: $K = 2.8 \times 10^5$–$3.0 \times 10^8$
3 Trotter steps: $K = 5.4 \times 10^5$–$1.0 \times 10^8$
$\propto 1/N$
monoprop
PauliPropagation.jl
ppvm
MajoranaPropagation.jl
ns per gate, per term
all $K$ terms (dense codes)
anticommuting (monoprop)
terms per gate
231×
65×
qubits $N$
modes $N$
32
64
128
256
512
1024